

A Semantic Knowledge-Driven and Explainable Framework for Customer Lifetime Value Prediction and VIP Targeting in Online Retail

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Abstract

Customer Lifetime Value (CLV) prediction in online retail faces two simultaneous pathologies: zero inflation (47–67% of customers have zero future spend) and heavy-tailed non-zero spending (skewness > 22). We propose a *semantic knowledge-driven* and explainable framework that extends classical temporal CLV modelling in three directions. First, we introduce **semantic product graph features**: a co-purchase graph encoded by Positive PMI and Truncated SVD yields customer taste profiles that are statistically orthogonal to RFM (Adjusted Rand Index **0.03**) and partition customers into six interpretable taste-based segments ($p=4.5 \times 10^{-5}$); these features enrich **interpretable taste segmentation** (24.5% of Stage-2 SHAP importance) and complement—rather than replace—RFM-based CLV ranking signals. Second, we compare 17 models on Online Retail II (805,549 transactions, 5,038 customers, 24 months) under walk-forward evaluation (three primary windows for the 17-model benchmark, all five windows for the semantic and CQR analyses), showing that a two-stage Hurdle Model achieves Normalized Gini **0.834** (95% bootstrap CI [0.767, 0.883]) and 60.96% Revenue Capture at top 10%—comparable to recent deep CLV baselines—while remaining fully interpretable. Third, we apply Conformalized Quantile Regression (CQR) to the Hurdle Model, achieving 95.2% empirical coverage at the nominal 95% level. Uncorrected MC-Dropout intervals from MCD-ZILN cover only 34.2% of test values; split-conformal widening restores MCD coverage to 94.7%, confirming that heavy-tailed CLV requires explicit calibration rather than nominal Bayesian intervals alone. SHAP analysis identifies Monetary, M/T ratio and recent spending as dominant predictors, with semantic dimensions accounting for 24.5% of Stage-2 importance.

Keywords: Customer lifetime value, VIP targeting, Hurdle model, Explainable AI, Semantic intelligence, Product graph embeddings, Conformalized quantile regression, Walk-forward cross-validation, Zero-inflated distribution, Online retail analytics

1 Introduction

Customer Lifetime Value (CLV) prediction directly informs marketing budget allocation, retention strategy, and personalised engagement [1, 2]: accurately identifying high-value customers lets retailers focus scarce resources on the segments most likely to generate future revenue. Throughout this paper, RFM acts as the descriptive segmentation lens, predicted CLV as the ranking target, and walk-forward Revenue Capture@ $K\%$ as retention-style validation that top-ranked customers actually return revenue in held-out windows. The task is challenging because CLV is simultaneously *zero-inflated* (47–67% of customers have zero future spend in our windows) and *heavy-tailed* (skewness > 22), violating MSE-regression assumptions and demanding specialised approaches [3].

Classical CLV models (BG/NBD + Gamma-Gamma) [4, 5] rely on summary RFM features; deep approaches such as ZILN [3], OptDist [6], dRNN [7], and MCD-ZILN [8] vary in interpretability and calibration. Three gaps persist: (i) few comparisons use leakage-free temporal evaluation; (ii) the marginal value of sequence and semantic features over tuned tabular methods is unclear; (iii) nominal 95% uncertainty intervals from MC Dropout cover only $\sim 34.2\%$ of test CLV on our benchmark. This paper closes those gaps with six contributions:

1. A **time-aware evaluation framework**: walk-forward CV across five windows (9–21 months observation), bootstrap 95% CIs and paired t -tests.
2. A **semantic product graph** representation: a co-purchase graph encoded by Positive PMI and Truncated SVD yields 32-dim product embeddings; per-customer revenue-weighted profiles capture latent product-taste knowledge orthogonal to aggregate RFM, with K -means clusters that are statistically distinct in CLV ($p=4.5 \times 10^{-5}$) and only weakly aligned with RFM clusters (ARI 0.03).
3. A **comprehensive comparison of 17 models** on Online Retail II spanning seven families (baselines, probabilistic, linear, gradient boosting, Hurdle, ZILN-family, sequence DL).
4. An **ablation study** of feature groups plus **two-stage SHAP explainability** for both Hurdle stages.
5. **Supervised dimension selection** and **recency-aware semantic profiling** that contribute 24.5% of Stage-2 SHAP importance and yield interpretable taste clusters orthogonal to RFM (ARI 0.03), without statistically significant aggregate NG gains ($p \geq 0.35$).
6. **Conformalized Quantile Regression (CQR)** [16] on the Hurdle Model, achieving 95.2% empirical coverage; split-conformal calibration of MCD-ZILN

raises its coverage from 34.2% to 94.7%, isolating calibration—not architecture—as the key uncertainty bottleneck.

This work advances *semantic intelligence* in retail analytics by constructing product co-purchase graph embeddings that capture latent customer taste profiles orthogonal to traditional RFM features. The remainder of the paper covers related work (Sec. 2), methodology (Sec. 3), experimental setup (Sec. 4), results (Sec. 5), and discussion plus conclusion (Sec. 6–7).

2 Related Work

Classical CLV models. BG/NBD [4] and Gamma-Gamma [5] anchor non-contractual CLV modelling but assume permanent churn and frequency–value independence.

Machine learning and deep CLV. Tree ensembles with Optuna tuning [10] remain strong on tabular data [15]. Deep CLV methods include ZILN loss [3], OptDist [6], dRNN [7], and MCD-ZILN [8]. Few prior studies jointly address temporal leakage, fair deep-vs-tree comparison, semantic product knowledge, and calibrated uncertainty.

Semantic intelligence. Product co-purchase graphs form a knowledge layer over transactions [17–19]. We treat PPMI–SVD embeddings as semantic profiles orthogonal to RFM, evaluated under walk-forward protocol with two-stage SHAP.

Hurdle models and explainability. Hurdle architectures [11, 12] decouple purchase incidence from conditional spend. SHAP [13] and Normalized Gini [3] support interpretability and ranking evaluation respectively.

3 Methodology

The proposed framework comprises data cleaning, feature engineering, walk-forward CV, model training, evaluation, VIP simulation, and two-stage SHAP analysis.

3.1 Problem Formulation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ denote the dataset, where $\mathbf{x}_i \in \mathbb{R}^d$ is a feature vector summarising customer i ’s observation-window history, and $y_i \in \mathbb{R}_{\geq 0}$ is the actual revenue generated by customer i during a future prediction window. The CLV prediction task is to learn $\hat{f} : \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$ minimising a chosen loss while preserving customer ranking quality.

A central challenge is that $y_i = 0$ for a large fraction of customers (non-returning buyers), while the non-zero values are heavy-tailed. We thus formulate CLV as a mixture distribution:

$$P(Y = y) = \begin{cases} 1 - \pi, & y = 0, \\ \pi f_Y(y; \theta), & y > 0, \end{cases} \quad (1)$$

where $\pi = P(Y > 0)$ is the purchase probability and $f_Y(y; \theta)$ is the conditional density of positive values. Both the Hurdle Model and the ZILN-family methods are instances of Eq. (1) that differ in their parameterisation and training procedure.

3.2 Temporal Walk-Forward Cross-Validation

We use five non-overlapping walk-forward windows W1–W5, all anchored at 2009-12-01, with observation/prediction horizons (in months): W1 = 12/3, W2 = 15/3, W3 = 18/6 (the primary 6-month-horizon window), W4 = 9/3 (shortest history), and W5 = 21/3 (longest history). Window sizes range from 3,313 to 5,245 customers and zero-rates from 42% to 67%. Within each window we apply a stratified 80/20 train/test split preserving the VIP class balance (top 10% by actual CLV).

3.3 Feature Engineering

We construct 23 base features in four groups: **RFM** (Recency, Frequency, Monetary; 3); **Behavioural** (Tenure, ActiveMonths, ProductDiversity, AvgOrderValue, Avg-DaysBetweenOrders, Regularity, IsUK; 7); **Interaction** (log-Monetary, log-Frequency, log-AOV, M/F, M/T, ActiveMonths/Tenure; 6); and **Monthly sequence summaries** (mean, std, max, recent-3 mean and max, active-month count, trend; 7). A fifth group, **Semantic product-graph features** (Sec. 3.7), adds 8 supervised top-correlation dimensions from the full profile plus 8 from a 90-day recency-aware profile, a drift scalar, and product counts.

3.4 Proposed Two-Stage Hurdle Model

Why a Hurdle architecture rather than a single XGBoost? A single MSE-trained regressor must optimise two conflicting objectives—pulling predictions toward zero for the 47–67% no-purchase mass while tracking long-tail VIPs—under one loss. The Hurdle decouples them, training one classifier on *whether* a customer returns and one regressor on *how much* a returner spends, each with the loss appropriate to its sub-task. The single-loss baselines in Sec. 5 therefore act as ablations of components the Hurdle subsumes.

We decompose Eq. (1) into two LightGBM [9] stages: a binary **classifier** estimating $\hat{\pi}_i = P(y_i > 0 \mid \mathbf{x}_i)$ on all N customers, and a **regressor** estimating $\hat{m}_i = \mathbb{E}[\log(1+y_i) \mid y_i > 0, \mathbf{x}_i]$ on the positive subset. The final prediction combines both with a lognormal bias correction:

$$\hat{y}_i = \hat{\pi}_i \left(e^{\hat{m}_i} - 1 \right) \exp\left(\frac{\hat{\sigma}^2}{2}\right), \quad (2)$$

with $\hat{\sigma}^2$ the validation residual variance of the log-target on positive customers (a homogeneous lognormal correction; heteroscedastic Stage-2 modelling is left to future work); both stages use early stopping.

3.5 Comparison Models

We compare the Hurdle Model against 16 alternatives: simple baselines (Mean, Monetary, RFM), linear models, BG/NBD + Gamma-Gamma, Optuna-tuned XGBoost/LightGBM, ZILN-family deep models (ZILN, OptDist, MCD-ZILN), and dRNN sequence DL.

3.6 Evaluation Metrics

Regression metrics: MAE, RMSE, R^2 . Ranking metrics: Spearman rank correlation and the Normalized Gini coefficient $\text{NG}(y, \hat{y}) = G(y, \hat{y})/G(y, y)$ [3] (1 is perfect, 0 is random). For business value, Revenue Capture at top $K\%$ is $\text{RC@}K\% = (\sum_{i \in \mathcal{T}_K} y_i) / (\sum_{i=1}^N y_i)$ where $\mathcal{T}_K$ are the top- $K\%$ customers by predicted CLV; lift is $\text{RC@}K\% / K\%$ and is reported as a linear rescaling of RC, not as an independent metric. We additionally report decile MAPE for calibration and top-5% MAPE for VIP accuracy. **NG and RC@10% are the primary headline metrics for model selection** (matching the marketing-budget use case); MAE/RMSE are secondary calibration checks; Spearman, R^2 and decile MAPE are diagnostic.

3.7 Semantic Product Graph Features

Two customers with identical RFM profiles may exhibit very different future behaviour depending on *what* they buy (e.g. kitchenware vs. seasonal gifts). We capture this latent product-taste knowledge through a co-purchase graph of products appearing in at least five distinct invoices ($|\mathcal{V}|=3,772$). Pair co-occurrence counts c_{ij} are converted to Positive PMI [17]:

$$\text{PPMI}(p_i, p_j) = \max\left(0, \log \frac{c_{ij} N_{\text{inv}}}{f_i f_j}\right), \quad (3)$$

with f_i the number of invoices containing p_i . Truncated SVD with $k=32$ produces an ℓ_2 -normalised product embedding matrix $\mathbf{E} = \mathbf{U} \mathbf{\Sigma}^{1/2} \in \mathbb{R}^{|\mathcal{V}| \times 32}$ (equivalent to skip-gram factorisation [18] and used in co-purchase recommendation [19]). Embeddings are learned *solely on the observation portion* of each window to prevent leakage.

For customer i we compute two revenue-weighted semantic profiles—a full-period $\mathbf{s}_i^{\text{full}}$ and a 90-day recency-aware $\mathbf{s}_i^{\text{recent}}$:

$$\mathbf{s}_i^{\text{full}} = \frac{\sum_j a_{ij} \mathbf{e}_{p_{ij}}}{\sum_j a_{ij}}, \quad \mathbf{s}_i^{\text{recent}} = \frac{\sum_{j: t_{ij} \geq T_{\text{end}} - 90\text{d}} a_{ij} \mathbf{e}_{p_{ij}}}{\sum_{j: t_{ij} \geq T_{\text{end}} - 90\text{d}} a_{ij}}, \quad (4)$$

plus a taste-drift scalar $d_i = 1 - \cos(\mathbf{s}_i^{\text{full}}, \mathbf{s}_i^{\text{recent}})$. Rather than unsupervised PCA, we select the top- $K=8$ dimensions per profile by $|\rho_{\text{Spearman}}(\cdot, \log(1+y))|$ on the training fold only. Combined with d_i and per-window product counts this yields 19 semantic features. We evaluate four ablation variants: HURDLE-RFM (16 features), HURDLE-SEQ (+7 sequence, 23), HURDLE-SEMV2 (+19 semantic, 35), and HURDLE-ALLV2 (+Sem $\times$ Seq interaction, 43).

3.8 Statistical Testing

For the main 17-model benchmark, paired t -tests are conducted across the three primary walk-forward windows and interpreted cautiously due to limited statistical power. For the semantic feature variants and the CQR analyses, paired comparisons use all five windows. Stage-1 AUC is always computed on the held-out test fold. All tests are complemented with bootstrap 95% CIs (1,000 test-set resamples per window).

Table 1 Hyperparameter tuning budget: GBM vs. deep CLV baselines.

	GBM	Deep
Search	Optuna (20 trials)	Fixed config
LR / dropout	Tuned	10^{-3} / 0.3–0.4
Pilot Optuna (W3 ZILN)	—	10 trials

4 Experimental Setup

4.1 Dataset

We use the Online Retail II dataset from the UCI Machine Learning Repository [14], containing transaction records from a UK-based non-store online retailer between December 2009 and December 2011. After preprocessing, 805,549 transactions remain across 5,038 customers (Dec 2009–Dec 2011; 41 countries; total revenue \$17.7M; median customer spend \$899). Window 1/3 zero-rates are 67.3%/47.3% with CLV skewness 22.95 on W3.

4.2 Implementation Details

Implemented in Python 3.12 (scikit-learn 1.5, LightGBM 4.5, XGBoost 2.1, PyTorch 2.12, Optuna 3.6, lifetimes 0.11.3). Random seed 42 throughout. Gradient-boosting models use Optuna with 20 trials, 3-fold internal CV and early stopping; deep models use fixed Adam ($\text{lr} = 10^{-3}$, $\text{wd} = 10^{-5}$), batch size 64, gradient clipping at 1.0, up to 200 epochs with early stopping—a deliberate asymmetry discussed in Sec. 6. Table 1 lists the fixed deep-learning configuration; GBM search spaces are in the public code. The ZILN loss clips $\mu \in [-10, 15]$ and $\sigma \in [0.01, 3.0]$ via softplus.

5 Results and Analysis

5.1 Main Model Comparison

Table 2 reports mean performance across the three primary walk-forward windows W1–W3, for which all 17 model families were fully executed; semantic variants (Sec. 5.6) and CQR (Sec. 5.7) are additionally evaluated on W4–W5. Bold marks the best per column; proposed methods are underlined. The proposed Hurdle Model achieves the highest Revenue Capture at top 10% (60.96%) while MCD-ZILN marginally leads mean Normalized Gini (0.834); six top methods cluster in a narrow band (NG 0.827–0.834).

The Monetary baseline (rank by historical spend) achieves NG 0.822 and RC@10% 60.02%—only -0.012 NG and -0.94 percentage points behind the best learned method. The top five methods improve RC@10% by less than 1 pp over Monetary; the proposed Hurdle Model leads with $\Delta\text{RC@10\%} = +0.94$ pp and $\Delta\text{MAE} = -\$73$. On retail data with strong spending inertia, advanced models therefore deliver consistent but modest gains; their main value is robustness across regimes and SHAP-enabled interpretability rather than large absolute uplift.

Table 2 Performance of 17 CLV Prediction Models (mean across three walk-forward windows). Proposed methods are underlined. dRNN mean MAE is dominated by Window 1 (\$16,772; $R^2=-155$); see Table 3.

Rank	Model	MAE (\$)	NG	RC@10%
1	MCD-ZILN [8]	732	0.834	60.68%
2	<u>Hurdle Model (Proposed)</u>	449	0.834	60.96%
3	ZILN [3]	618	0.833	60.20%
4	<u>Hurdle + Sequence</u>	457	0.833	59.58%
5	<u>OptDist [6]</u>	684	0.833	60.11%
6	dRNN [7]	6,075	0.827	60.23%
7	XGBoost (Optuna)	450	0.822	58.85%
8	Monetary baseline	522	0.822	60.02%
9	XGBoost (log target)	480	0.816	56.85%
10	LightGBM + Sequence	478	0.811	56.65%
11	LightGBM (Optuna)	472	0.810	58.80%
12	Linear Regression	456	0.810	60.55%
13	Ridge (log)	776	0.797	55.15%
14	LightGBM (log)	478	0.794	56.12%
15	RFM Score	707	0.716	52.68%
16	BG/NBD + GG [4, 5]	550	0.485	3.23%
17	Mean Predictor	765	-0.049	9.72%

Table 3 dRNN per-window calibration vs. Hurdle MAE (W1-W3).

W	Zero (%)	MAE	NG	RC@10%	Hurdle
1	67.3	16,772	0.837	66.0%	380
2	42.0	299	0.765	49.4%	250
3	47.3	1,154	0.879	65.3%	718

5.2 Bootstrap CIs and Statistical Significance

Bootstrap 95% CIs (1,000 resamples) for the Hurdle Model give $NG = 0.834$ [0.767, 0.883] and $RC@10\% = 60.96\%$ [48.87, 69.71]. Paired t -tests against MCD-ZILN are significant ($p < 0.05$) only against the six weakest models; the remaining ten methods (Hurdle $p=0.27$, ZILN $p=0.78$, dRNN $p=0.12$) show no statistically significant difference, indicating an apparent $NG \approx 0.83$ ceiling (Sec. 6).

5.3 VIP Targeting and Profit Simulation

At $K=10\%$, random targeting captures 11.8% of revenue, the oracle 72.2%, and the Hurdle Model 60.8% ($\eta=81.2\%$ oracle efficiency). A Window-3 profit simulation (20% margin, \$3/contact) yields \$292,314 net profit for Hurdle vs. \$47,706 random (+513%).

5.4 Feature Group Ablation Study

Fig. 1 reports the ablation across three windows. Starting from RFM-only (NG 0.822), behavioural features add +0.010, interactions add +0.008, and monthly-sequence features add the largest single gain (+0.015, NG 0.836). Although sequence features

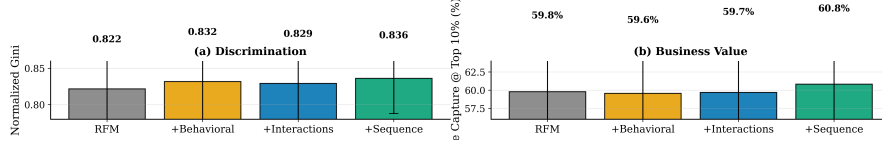


Fig. 1 Feature group ablation study across all three walk-forward windows (Hurdle Model with Optuna-tuned LightGBM). Error bars show window-to-window variation. Adding monthly sequence features yields the single largest mean improvement of +0.015 Normalized Gini over the RFM-only baseline.

provide a modest standalone NG uplift, they account for $\sim 24\%$ of total feature importance (Sec. 5.5).

Stage-wise diagnostics. On the stratified 80/20 test fold across W1–W3, the Stage-1 classifier averages AUC-ROC 0.798 ± 0.018 and AUC-PR 0.759 ± 0.074 (all metrics on the held-out test set, fixed LightGBM hyperparameters). Semantic-variant S1-AUC values in Table 4 use Optuna-tuned LightGBM over five windows and therefore differ slightly (0.787–0.791 vs. 0.798); both are test-fold scores, not validation-fold estimates. The Stage-2 regressor achieves $R^2 = 0.407 \pm 0.106$ and MAE $\$776 \pm 334$ on the positive subset. On Window 3, the Hurdle top-decile captures 65.07% of actual revenue versus 0.12% in the bottom decile ($524\times$ discrimination).

5.5 Explainability and Walk-Forward Stability

Monetary leads permutation importance (0.037), followed by M/T and `seq_recent3_mean`. SHAP: Stage 1 is driven by ActiveMonths ($|\text{SHAP}|=0.291$), Recency (0.255), and `seq_recent3_mean` (0.197); Stage 2 by Monetary (0.365) and M/T (0.155). All advanced models maintain NG > 0.77 across five windows.

5.6 Semantic Product Graph Feature Analysis

Two refinements support semantic analysis: **supervised dimension selection** (top- $K=8$ dims by training-fold Spearman correlation) and **recency-aware profiling** (full-period and 90-day profiles plus drift; Sec. 3.7).

HURDLE-SEMV2 achieves the highest mean RC@10% (57.00%). Paired t -tests against HURDLE-RFM yield $p \geq 0.35$ for NG, RC@10% and test-fold Stage-1 AUC—semantic features do not produce statistically significant aggregate ranking gains at $n=5$.

Interpretability. SHAP on HURDLE-ALLV2 (W3) shows semantic features account for 24.5% of Stage-2 importance; six K -means taste clusters are near-orthogonal to RFM (ARI 0.032) with mean CLV $\$484$ – $\$1,531$ ($p=4.5 \times 10^{-5}$).

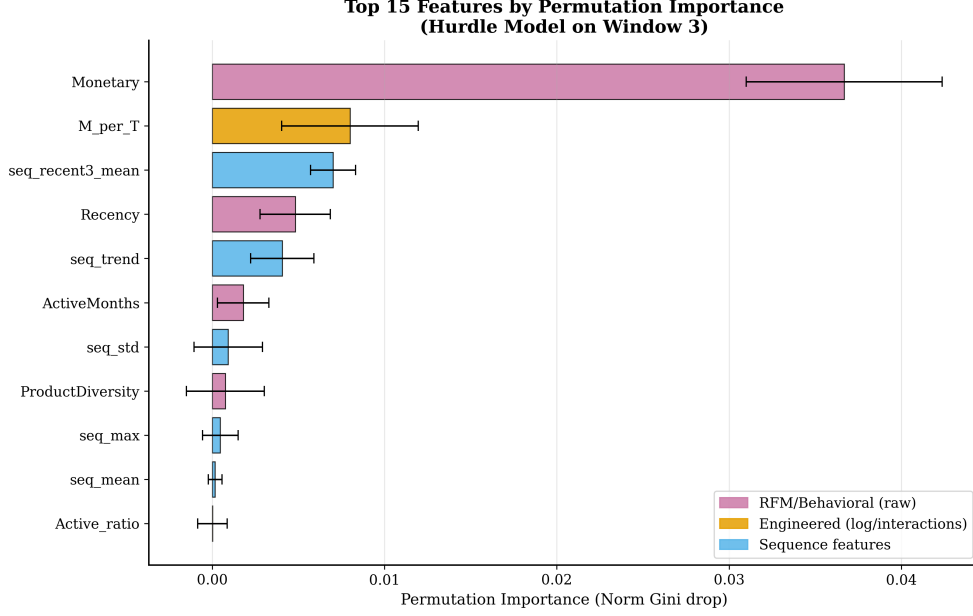


Fig. 2 Top 15 features by permutation importance for the Hurdle Model on Window 3. Monetary leads (0.037), followed by M/T and `seq_recent3_mean`; sequence features cumulatively account for $\sim 24\%$ of total importance.

Table 4 Hurdle Model Variants Across Five Walk-Forward Windows (mean $\pm$ std, $n=5$). SEMV2 = supervised top-8 + recency-aware features (19 features). Bold marks the best per column. S1-AUC is Stage-1 AUC-ROC on the stratified 80/20 *test* fold with Optuna-tuned LightGBM (15 trials); fixed-hyperparameter diagnostics on W1–W3 yield mean 0.798 ± 0.018 (Sec. 5).

Variant	N	NG	RC@10%	S1-AUC (test)
Hurdle-RFM	16	0.794$\pm$0.067	56.82 $\pm$ 8.9	0.787
Hurdle-Seq	23	0.788 $\pm$ 0.079	56.70 $\pm$ 9.1	0.789
Hurdle-SemV2	35	0.791 $\pm$ 0.074	57.00$\pm$9.0	0.791
Hurdle-AllV2	43	0.782 $\pm$ 0.083	56.70 $\pm$ 8.9	0.782

5.7 Calibrated Prediction Intervals

MCD-ZILN’s nominal 95% MC-Dropout intervals cover only $34.2\% \pm 9.6\%$ of test CLV values under heavy tails. Applying the same split-conformal widening used by CQR raises MCD coverage to $94.7\% \pm 0.8\%$, showing that *calibration*, not model family, is the primary bottleneck. We therefore apply Conformalized Quantile Regression (CQR) [16] to the Hurdle Model ($\alpha=0.05$) for deployable intervals with provable marginal coverage.

CQR-Hurdle achieves **95.2% $\pm$ 0.7%** coverage with mean width \$1,607—3.6 $\times$ narrower than conformal-calibrated MCD (\$5,729) at comparable coverage. Raw quantile coverage alone reaches 94.8%; positive conformal correction $\hat{q}>0$ in three of

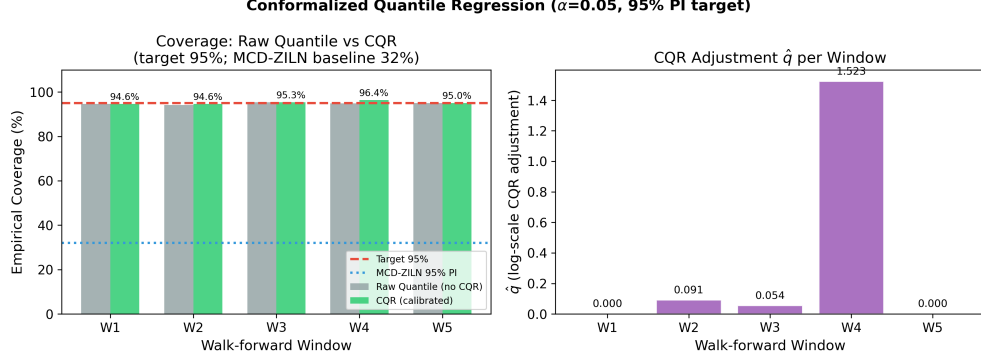


Fig. 3 Conformalized Quantile Regression (CQR) coverage at $\alpha=0.05$ (95% target) across five walk-forward windows. Left: raw-quantile vs. conformal-adjusted coverage, with the 95% target. Raw MCD MC-Dropout coverage (34.2%) and split-conformal MCD (94.7%) are shown for reference (Table 5).

Table 5 CQR vs. conformal-calibrated MCD at $\alpha=0.05$ (95% target). All coverages are test-fold empirical values.

Method	Raw (%)	Conf. (%)	Width (\$)
MCD-ZILN (MC Dropout)	34.2	94.7	5,729
CQR-Hurdle (quantile)	94.8	95.2	1,607

five windows. Midpoint ranking under CQR yields NG 0.792 ± 0.057 and RC@10% $55.29\% \pm 10.1$.

6 Discussion

Convergence and method selection. Six top methods cluster at NG 0.827–0.834 ($p \geq 0.12$). Method choice should follow secondary criteria: interpretability favours Hurdle; deployable uncertainty favours CQR-Hurdle; sequence-rich regimes favour dRNN/OptDist. Tree boosters match deep models at this scale [15].

Hyperparameter tuning asymmetry. Gradient-boosting models received Optuna search (20 trials, 3-fold CV); deep baselines used fixed Adam settings (Table 1). We retain this *deliberate asymmetry* because deep training is 10–30 $\times$ slower at $N \approx 5,000$ and our focus is an interpretable tree-first benchmark, not deep SOTA. We treat this deliberate asymmetry as a feature of the study rather than a limitation, as it reflects realistic scenarios where tree-based models are more practical to tune on retail-scale data. We do *not* interpret comparable NG/RC@10% as architectural inferiority. A pilot 10-trial Optuna on ZILN (W3) yields test NG 0.884 vs. 0.885 fixed—negligible and insufficient to break the NG ≈ 0.83 ceiling.

dRNN dollar-scale calibration. Table 3 shows dRNN mean MAE (\$6,075) is driven by W1 (\$16,772; $R^2 = -155$); W2–W3 MAE is \$299–\$1,154 with NG 0.765–0.879. For VIP targeting we prioritise NG and RC@10% over aggregate MAE for sequence models.

Monetary baseline. Historical-spend ranking achieves NG 0.822 and RC@10% 60.02%—only -0.012 NG behind the best learned method. The Hurdle Model adds consistent MAE gains, two-stage SHAP, and semantic taste clusters the Monetary baseline cannot produce.

Limitations. (i) The study uses a single UK retailer dataset ($N=5,038$). While this is a standard benchmark, results may not fully generalize to other retail verticals or emerging markets; cross-retailer validation is planned as future work. (ii) Five walk-forward folds limit statistical power. (iii) Deep baselines used fixed hyperparameters, while GBM models received full Optuna tuning (deliberate asymmetry discussed in Sec. 6). (iv) Stage-2 uses a homogeneous $\hat{\sigma}^2$ correction. (v) CQR and conformal MCD provide marginal, not conditional, coverage. (vi) Semantic predictive gains are not significant at $n=5$; value is primarily interpretability and taste segmentation.

7 Conclusion and Future Work

We presented a semantic knowledge-driven framework for CLV prediction and VIP targeting on Online Retail II. A two-stage Hurdle Model achieves NG 0.834 and RC@10% 60.96% under walk-forward evaluation, comparable to recent deep baselines while remaining interpretable. Semantic graph features yield taste clusters orthogonal to RFM (ARI 0.03) and 24.5% Stage-2 SHAP importance, but not statistically significant aggregate ranking gains ($p \geq 0.35$). CQR delivers 95.2% coverage with \$1,607 mean width; split-conformal calibration raises MCD-ZILN from 34.2% to 94.7% coverage, isolating calibration as the key uncertainty issue. Future work will add deep hyperparameter parity across all windows, heteroscedastic Stage-2 modelling, and cross-retailer validation.

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Data and Code Availability. The Online Retail II dataset is publicly available at <https://archive.ics.uci.edu/dataset/502/online+retail+ii>. All implementation code and preprocessing scripts will be released in a public repository upon acceptance, and are available from the author on request in the meantime.

Conflict of Interest. The author declares none.

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